Querry 1: Filter all transactions that occurred in the year 2023. (Using Where clause)

SQL Querry

```
-----Querry 1 filter all transaction that occured in the year 2023
SELECT
  *
FROM
  `my-practice-project-493417.Retail_Sales.Retail`
WHERE
  EXTRACT(YEAR FROM Date) = 2023
LIMIT 1000;
```

Table results (all columns had to aggregate the row)

Job information | Results | Visualization | JSON | Execution details | Execution graph

row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25

Question 2: Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.

SQL Querry

```
----Querry 2  Filtering + Conditions .
SELECT ...
FROM
  `my-practice-project-493417.Retail_Sales.Retail`
QUALIFY `Total Amount` > AVG(`Total Amount`) OVER();
```

Table results ,

row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	500
8	537	2023-06-03	CUST537	Female	21	Beauty	1	500	500

Question 3 . Calculate the total revenue (sum of Total Amount).

SQL query

Untitled query

Run Save Download Sh

```
SELECT
  SUM(Retail.`Total Amount`) AS total_revenue
FROM
  `my-practice-project-493417.Retail_Sales.Retail` AS Retail
```

Results Table

Query results

Job information | Results | Visualization | JSON | Execution details | Execution grap

Row	total_revenue
1	456000

Query 4

```
----query: Display all distinct Product Categories in the dataset.

select
distinct `Product Category`
FROM
`my-practice-project-493417.Retail_Sales.Retail`
```

Table result

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Question 5: For each Product Category, calculate the total quantity sold.

SQL queries

```
----Query 5-for each Product Category, calculate the total quantity sold.
SELECT
    Retail.`Product Category`,
    SUM(Retail.Quantity) AS total_quantity
FROM
    `my-practice-project-493417.Retail_Sales.Retail` AS Retail
GROUP BY
    Retail.`Product Category`;
```

Row	Product Category	total_quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Query 6: Create a column called Age Group that classifies customers as: – 'Youth' (<30)

```
----Query 6 Case statement,Create a column called Age_Group that classifies customers as: - 'Youth' (<30)
SELECT *,
  CASE
    WHEN Age < 30 THEN 'Youth'
    ELSE 'Adult'
  END AS Age_Group
  FROM `my-practice-project-493417.Retail_Sales.Retail`;
```

Table results

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount	Age_Group
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25	Adult
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25	Adult
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25	Adult
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25	Adult
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25	Youth
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25	Youth
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25	Adult
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25	Youth
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	25	Adult
10	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25	Adult
11	512	2023-11-07	CUST512	Female	57	Beauty	1	25	25	Adult
12	739	2023-11-29	CUST739	Male	36	Beauty	1	25	25	Adult

Query 7: For each Product Category, show only those categories where the total revenue exceeds 5,000.

SQL queries,

```
SELECT
    retail.Gender,
    COUNTIF(retail.`Total Amount` > 500) AS high_value_transaction_count
  FROM
    `my-practice-project-493417.Retail_Sales.Retail` AS retail
  GROUP BY
    retail.Gender;
```

Tables results

	Gender	high_value_transaction_count
1	Male	144
2	Female	155

```
SELECT
    Retail.`Transaction ID`,
    Retail.Date,
    Retail.`Customer ID`,
    Retail.Gender,
    Retail.Age,
    Retail.`Product Category`,
    Retail.Quantity,
    Retail.`Price per Unit`,
    Retail.`Total Amount`,
    CASE
        WHEN Retail.`Price per Unit` < 50 THEN 'Cheap'
        WHEN Retail.`Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
        WHEN Retail.`Price per Unit` > 200 THEN 'Expensive'
    END AS Unit_Cost_Category
FROM
    `my-practice-project-493417.Retail_Sales.Retail` AS Retail
LIMIT 10;
```

Tables results

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount	Unit_Cost_Categ
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25	Cheap
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25	Cheap
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25	Cheap
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25	Cheap
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25	Cheap
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25	Cheap
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25	Cheap
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25	Cheap

SQL queries

Combining WHERE + CASE: Display all transactions from customers aged 40 or older and add a column Spending Level showing: – 'High' if Total Amount > 1000 – 'Low' otherwise

SQL queries

```
SELECT
    retail.`Transaction ID`,
    retail.`Date`,
    retail.`Customer ID`,
    retail.`Gender`,
    retail.`Age`,
    retail.`Product Category`,
    retail.`Quantity`,
    retail.`Price per Unit`,
    retail.`Total Amount`,
    CASE
        WHEN retail.`Total Amount` > 1000 THEN 'High'
        ELSE 'Low'
    END AS Spending_Level
FROM
    `my-practice-project-493417.Retail_Sales.Retail` AS retail
WHERE
    retail.Age >= 40;
```

Tables results

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount	Spending_Level
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25	Low
2	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25	Low
3	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25	Low
4	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25	Low
5	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25	Low
6	512	2023-11-07	CUST512	Female	57	Beauty	1	25	25	Low
7	791	2023-12-05	CUST791	Female	51	Beauty	1	25	25	Low
8	825	2023-08-26	CUST825	Female	46	Beauty	1	25	25	Low